Photovoltaic panels convert sunlight directly into electricity using semiconductor cells. A typical residential panel has an efficiency of about 20 percent, and most manufacturers provide a performance warranty of 25 years. Silicon is the most widely used cell material.

Installation requires a roof that faces south in the northern hemisphere, with a tilt angle roughly equal to the local latitude. Shading from trees or chimneys can reduce output by up to 40 percent. An inverter converts the direct current from the panels into alternating current for household use.

Maintenance is minimal. Panels should be cleaned twice a year and inspected after storms. The inverter usually needs replacement after 10 to 15 years. Battery storage adds roughly 8000 dollars to the cost of a typical home system but allows use of solar power at night.